METHODS IN MOLECULAR BIOLOGY (MB504T)
CURRENT FINALTERM PAPER SPRING 2024(23July)

SHORT QUESTIONS(3)

1. Negative control and blank reaction

NEGATIVE CONTROL: Controls for specificity of the amplification reaction contains all reagents and a DNA template lacking the target sequence.

BLANK REACTION: Controls for contamination contains all reagents except DNA template.

2. dNTPs

Deoxynucleoside triphosphates (dNTPs) are the building blocks from which the DNA polymerase synthesizes a new DNA strand during successive cycles of PCR amplification.

dNTPs consist of four basic nucleotides - dATP, dCTP, dGTP, and dTTP. The concentration of each dNTP (dATP, dCTP, dGTP, dTTP) in the PCR is usually 200 μM.

3. Give the name of six PCRs

- Standard PCR
- Real-Time PCR
- Reverse Transcription PCR
- Nested PCR
- Multiplex PCR
- Hot Start PCR
- Long Range PCR
- Assembly PCR
- Allele-Specific PCR
- Colony PCR

4. What is Cell line?

Cell lines are cultures of animal cells that can be propagated repeatedly and sometimes indefinitely. They arise from primary cell cultures. Primary cultures are initiated directly from the cells, tissues, or organs of animals and are typically used in experiments within a few days. Cell lines are frozen to provide a source of inoculum for future cultures

5. Assembly PCR

Assembly PCR used to assemble two or more pieces of DNA into one piece.

- **Assembly PCR Principle**:

Assembly PCR is the synthesis of long DNA structures by performing PCR on a pool of long oligonucleotides with short overlapping segments, to assemble two or more pieces of DNA into one piece.

6. Which type of PCR used in amplification of DNA from fossils and how false amplification can be avoided/ Suicide PCR

Suicide PCR is typically used in paleogenetics or other studies where avoiding false positives and ensuring the specificity of the amplified fragment is the highest priority. The method prescribes the use of any primer combination only once in a PCR, which should never have been used in any positive control PCR reaction. Primers should always target a genomic region never amplified before in the lab using this or any other set of primers. This ensures that no contaminating DNA from previous PCR reactions is present in the lab, which could otherwise generate false positives. Use of any primers combination only once in a PCR.

7. SIGNIFICANCE OF SNPs

- In disease diagnosis

- In Finding predisposition to diseases
- In drug discovery & development
- In drug responses
- Investigation of migration patterns

8. SNP types

The SNPs in the coding region of gene are divided into two types:

SYNONYMOUS SNPS(Silent SNPs): The synonymous SNPs do not change the amino acid sequence of protein or not affect the protein function.

NON-SYNONYMOUS SNPS: SNPs with single amino acid substitution in a protein sequence that may lead to missense mutation or nonsense mutation.

- ***Missense SNPs:*** Change the amino acid sequence of the protein, potentially altering protein function.
- ***Nonsense SNPs:*** Introduce a premature stop codon, leading to a truncated protein that is usually non-functional.

9. Characteristics of SNPs

SNP definition: A SNP is defined as a single base change in a DNA sequence that occurs in a significant proportion (more than 1 percent) of a large population. In human beings, 99.9 percent bases are same. Remaining 0.1 percent makes a person unique.

- SNPs are found in § coding and (mostly) noncoding regions.
- Occur with a very high frequency § about 1 in 1000 bases to 1 in 100 to 300 bases.
- The abundance of SNPs and the ease with which they can be measured make these genetic variations significant.
- SNPs close to particular gene acts as a marker for that gene.
- SNPs in coding regions may alter the protein structure made by that coding region.

10. Two enzyme and methods for vitro molecules joining

Mainly three methods are used for joining DNA in vitro

i. Joining covalently annealed cohesive ends by DNA ligase
ii. Joining blunt-ended fragments by DNA ligase from phage T4 infected E. coli
iii. The third utilizes the enzyme terminal deoxynucleotidyltransferase to synthesize homopolymeric 3’ single stranded tails at the ends of fragments.

ENZYME NAMES:

i. DNA Ligase
ii. Alkaline Phosphatase

11. DNA ligase

An enzyme that creates a phosphodiester bond between the 3’ end of one DNA segment and the 5’ of another. E. coli and phage T4 encode enzyme, DNA ligase, which seals single stranded nicks between adjacent nucleotides in a duplex DNA chain. DNA ligase can repair the nicks produced after the association of cohesive ends of DNA strands

12. Role of phosphate removal during DNA cohesive end joining

Alkaline Phosphatase:

An enzyme responsible for removing phosphate groups from many types of molecules including DNA. Treatment of linearized plasmid vector DNA with alkaline phosphatase to remove 5’- terminal phosphate groups will prevent recircularization and plasmid dimer formation.

13. Difference between adapter and linker

ADAPTORS: Chemically synthesized adaptor molecules which have a preformed cohesive end can be used to insert blunt ended foreign DNA into vector. **OR** Adapters are added to the ends of DNA fragments to provide specific sequences for ligation or PCR priming.

LINKER: In contrast, linkers are inserted between two DNA fragments to connect them, providing flexibility.

The ***main difference*** between linker and adaptor is that former having blunt ends while later have one cohesive end.

14. Which method is used for detect polymorphism of DNA

AFLP PCR : AFLP is a highly sensitive PCR-based method for detecting polymorphisms in DNA. ☐ AFLP can be also used for genotyping individuals for a large number of loci.

15. On what basis are proteins separated in native gel electrophoresis?/Making protein on native gel electrophoresis on the base of which factor? asa hi kuch question tha

In native gel electrophoresis, proteins are separated based on their native properties, including charge, size, and shape. The technique preserves the protein's natural structure, allowing for separation according to their inherent characteristics, with proteins having a higher charge, smaller size, or more compact shape migrating faster through the gel.

16. Uses/Applications of PCR in Clinical Microbiology.

- ***Identification of bacteria:*** ☐ Slow growing bacteria ☐ Bacterial antibiotic resistance genes.
- ***VIRUS***: almost all viruses identification and viral load.
- ***FUNGUS:*** ☐ Aspergillus spp ☐ Candida spp ☐ Plasmodium spp ☐ Trypanosoma spp ☐ Leishmania spp ☐ Babesia spp

17. Uses of PCR in forensic science.

Forensic science is the application of scientific procedures to solve criminal and legal matters. Molecular methods are used to established the filiations of a person or to obtain evidence from minimal samples of saliva, semen or other tissues

18. DNA profiling.

DNA profiling or DNA fingerprinting is a forensic technique used to identify individuals by characteristics of their DNA.

19. Western Blotting and uses/Applications

A technique used to detect the presence of a specific protein in a complex protein mixture.

Principle: Western blotting separates proteins by size, transfers them to a membrane, and detects specific proteins using antibodies, allowing for identification and quantification of individual proteins in a complex mixture.

❖ ***APPLICATIONS/USES:***

- Analysis Of IgG Fractions purified from human plasma.
- Diagnosis of HIV by ELISA, involves the western blotting technique. Western blotting technique is also used to Detect Some Forms Of Lyme Disease.
- Western blotting technique is used in Definitive Test For BSE, which is commonly know as ***Mad cow disease.***
- Confirmatory Test For Hepatitis-B involves western blotting technique.
- Western blotting test is used in the Analysis Of Biomarkers such as hormones, growth factors & cytokines.
- This technique is also employed in The Gene Expression Studies

20. Limitations in Western Blotting

- very delicate and time consuming process. A minute imbalance at any level of the procedure can skew the results of the entire process.
- Incorrect labeling of the protein can happen due to the reaction of secondary antibody.
- Cause erroneous in bands or no bands due to insufficient transfer.
- Well trained technicians are required for this technique. Primary antibody availability is crucial.
- It is just a semi-quantitative at best. Only an approx. estimation & not a precise measurement of molecular weight of the protein is possible

21. Western Blotting And Who It Use?

Western blotting is a laboratory technique used to detect and analyze proteins in a sample, commonly used by research scientists, clinical researchers, biotechnology companies, and forensic scientists to identify, quantify, and characterize proteins through gel electrophoresis, transfer, probing with antibodies, and visualization.

22. Difference between southern blotting and northern blotting

Southern Blotting: Southern blot is used to detect the presence of a particular piece of DNA in a sample by a molecular probe. Southern Blotting is named after its inventor, the British Biologist Edwin Southern (1975).

☐ Identify DNA sequence (gene) of interest. ☐ Identified DNA may be a small piece of DNA or a mutation

Northern Blotting: Northern blot is a method used for detection of a specific RNA sequence in RNA samples. Northern blot is used in many ways while studying RNA.

23. Northern Blot – Applications

- Study of gene expression in eukaryotic cells.
- To measure the amount & size of RNAs transcribed from eukaryotic genes.
- To estimate the abundance of RNAs.
- To equalize the amounts of RNA loaded into lanes of gels.
- Use of housekeeping gene (endogenous constitutively-expressed gene).
- Normalizing samples according to their content of mRNAs of the housekeeping gene.

24. Principle of Southern Blotting

Southern blotting detects specific DNA sequences by fragmenting DNA, separating it by size, transferring it to a membrane, and hybridizing it with a labeled probe. The probe binds to the target sequence, allowing researchers to identify and analyze specific DNA sequences, making it a valuable tool for understanding gene structure and expression.

25. Southern blotting and who cleavage the DNA?

Southern blotting is a laboratory technique used to detect specific DNA sequences in a sample. It involves cleaving the DNA with restriction enzymes, which cut the DNA at specific recognition sites, resulting in fragments of varying lengths. These fragments are then separated by gel electrophoresis, transferred to a membrane, and hybridized with a labeled probe to detect the desired DNA sequence.

26. Uses of Southern Blotting

Southern blotting is used for:

- Detecting specific DNA sequences
- Identifying gene mutations and variations
- Analyzing gene expression and regulation
- Studying genome organization and structure
- Diagnosing genetic disorders
- Forensic analysis and DNA fingerprinting
- Cloning and sequencing genes

27. Transfer methods of DNA to membrane in southern blotting

- Upward capillary transfer
- Downward capillary transfer
- Simultaneous transfer to two membranes
- Electrophoretic transfer
- Vacuum transfer

28. Ion-Exchange Chromatography and types.

Ion exchange chromatography is a separation based on charge. Ion-exchange chromatography preserves analyte molecules on the column based on ionic interactions

Use of Ion-Exchange Chromatography: Used for almost any kind of charged molecules Large proteins, small nucleotides & amino acids

TWO TYPE OF ION EXCHANGERS:

1) ***Cation Exchanger:*** Possess negatively charged groups and these will attract positively charged cations Also called acidic ion exchangers.
2) ***Anion Exchanger:*** Have positively charged groups that will attract negatively charged anions Also called basic ion exchangers

29. Immunofluorescent Microscopy

Uses antibodies conjugated to fluorescent markers to locate specific structures on specimens. It allows specific structures to be visualized by illuminating them with ultraviolet light. This Technique is used For looking at surface markers on eukaryotic cells. Another Use is As a whole-cell staining technique in bacteriology.

30. Which purpose is using Triple antibody Sandwich ELISA and another name used for?

Also known as ***indirect ELISA***

It is often used to identify antibodies in patient blood. This may be sign of infection.

System is used to test patient blood for the presence of hepatitis B virus (HBV) antibodies

31. SNP-Haplotype

A set of closely linked genetic markers present on one chromosome which tend to be inherited together (not easily separable by recombination).

The “Haplotype centric” approach combines the information of adjacent SNPs into composite multi locus haplotypes Ø Haplotypes are not only more informative but also capture the regional LD information, which is assumed to be robust and powerful Ø Association of haplotype frequencies with the presence of desired phenotypic frequencies in the population will help in utilizing the maximum potential of SNP as a marker.

LONG QUESTIONS(5)

32. Difference between Nested PCR and multiplex.

- **Nested PCR:**

In this type of PCR the Two pairs (instead of one pair) of PCR primers are used to amplify a fragment. First pair amplify a fragment similar to a standard PCR. Second pair of primers – nested primers (as they lie / are nested within the first fragment). Second pair of primers bind inside the first PCR product fragment to allow amplification of a second PCR product. Increases the specificity of DNA amplification, by reducing background due to non-specific amplification of DNA. Very low probability of nonspecific amplification.

- **Multiplex PCR:**

Multiplex PCR is a variant of PCR which enables simultaneous amplification of many targets of interest in one reaction by using more than one pair of primers. The use of multiple, unique primer sets within a single PCR reaction to produce amplicons of varying sizes specific to different DNA sequences.

33. Define whole genome amplification PCR & suicide PCR

- **Whole Genome Amplification PCR:**

Primers can be designed to be 'degenerate' - able to initiate replication from a large number of target locations. Whole genome amplification (WGA) is a group of procedures that allow amplification to occur at many locations in a genome.

- **Suicide PCR**

Suicide PCR is typically used in paleogenetics or other studies where avoiding false positives and ensuring the specificity of the amplified fragment is the highest priority. The method prescribes the use of any primer combination only once in a PCR, which should never have been used in any positive control PCR reaction.

34. Define in-situ PCR, colony PCR, and Asymmetrical PCR?

- **IN SITU PCR**

In Situ PCR (ISH) is a polymerase chain reaction that actually takes place inside the cell on a slide.
□ In situ PCR amplification can be performed on fixed tissue or cells .

❖ **COLONY PCR**

The screening of bacterial or yeast clones for correct ligation or plasmid products, and this type of PCR is called Colony PCR.

❖ **ASYMMETRIC PCR**

Asymmetric PCR is used to amplify one strand of the original DNA more than the other.

It is used in some types of sequencing and hybridization probing where having only one of the two complementary strands is ideal. PCR is carried out as usual, but with a great excess of one primers for the chosen strand. Due to the slow amplification later in the reaction after the limiting primer has been used up, extra cycles of PCR are required.

35. Asymmetric PCR & its uses/Applications

Asymmetric PCR is used to amplify one strand of the original DNA more than the other.

☐ It is used in some types of sequencing and hybridization probing where having only one of the two

complementary strands is ideal.

. It is also used to generate long primers and single-stranded DNA for incorporating cross-linking analogs into specific sites in a DNA probe.

36. Colony PCR

The screening of bacterial or yeast clones for correct ligation or plasmid products, and this type of PCR is called Colony PCR.

Methodology:

☐ Pick a bacterial colony with an autoclaved toothpick, swirl it into 25 µl of TE autoclaved dH2O in an microfuge tube. ☐ Heat the mix in a boiling water bath (90-100ºC) for 2 minutes ☐ Spin sample for 2 minutes high speed in centrifuge. ☐ Transfer 20 µl of the supernatant into a new microfuge tube. ☐ Take 1-2 µl of the supernatant as template in a 25 µl PCR standard PCR reaction.

37. Explain Methylation PCR

Methylation-specific PCR is used to identify patterns of DNA methylation at CpG islands in genomic DNA.

MSP used in quantitative PCR provides information about methylation state of a given CpG island. The Target DNA is first treated with sodium bisulfite, which converts unmethylated cytosine bases to uracil, which is complementary to adenosine in PCR primers. Two amplifications are then carried out on the bisulfite-treated DNA: One primer set anneals to DNA with cytosines (corresponding to methylated cytosine), and the other set anneals to DNA with uracil (corresponding to unmethylated cytosine).

38. What is the length of primer? Application of mini-primer PCR and name enzyme that use in it

☐ Mini Primer PCR uses a thermostable polymerase (S-Tbr) that can extend from short ***primers as short as 9 or 10 nucleotides.*** ☐ This method permits PCR targeting to smaller primer binding regions, and is used to amplify conserved DNA sequences, such as the 16S (or eukaryotic 18S) rRNA gene. ☐ PCR that can extend from short primers

The enzyme used in mini-primer PCR is a ***heat-stable DNA polymerase, such as Taq polymerase***. This enzyme is originally isolated from the thermophilic bacterium Thermus aquaticus.

39. Radioactive & Non Radioactive Probes Characteristics

RADIOACTIVE PROBE - P32

- Sensitive
- Relatively cheap
- Hazardous
- Radioactive waste disposal regulations should be followed

NON-RADIOACTIVE PROBE - BIOTIN

- Sensitive
- Relatively expensive

40. Define Antibody Avidity, Adjuvant, Antibody Affinity

ANTIBODY AFFINITY: Antibody affinity is defined as strength of the binding interaction between antigen and antibody.

ANTIBODY AVIDITY: Antibody avidity describes the strength with which an antibody binds to a complex antigen.

ADJUVANTS: Adjuvants Substances which increase the immunogenicity of the antigen. These are used to reduce the amount of antigen required. ***Example of adjuvants:*** Adjuvants may be chemicals (detergents and oils) Complex proprietary products (containing bacterial cell walls).

41. Application of DNA fingerprinting

- ***Diagnosis and Developing cures for inherited disorders :***

• DNA fingerprinting is used to diagnose inherited disorders in both prenatal and newborn babies in hospitals around the world. • These disorders may include cystic fibrosis, hemophilia, Huntington's disease, familial Alzheimer's, sickle cell anemia, thalassemia, and many others. • Early detection of such disorders enables the medical staff to prepare themselves and the parents for proper treatment of the child. • In some programs, genetic counselors use DNA fingerprint information to help prospective parents □ In other programs, prospective parents use DNA fingerprint information in their decisions concerning affected pregnancies.

- ***Biological Evidence to Identify Criminals:***

• Where fingerprints are not available but biological specimens are available like blood or semen stains, hair, or items of clothing at the scene of the crime then these items may prove to be valuable sources of DNA of the criminal. • Since the year 1987, innumerable cases have been solved with the help of DNA fingerprint evidence.

- ***Paternity disputes :***

• Another important use of DNA fingerprints in the court system is to establish paternity in custody and child support litigation. In these applications, DNA fingerprints bring an unprecedented, nearly perfect accuracy to the determination.

- ***Personal Identification :***

• DNA maybe the best way to identify a person as all body tissues and organs contain the same DNA type. The specimen required also is very small. In fact the US army has been doing DNA fingerprinting of all its soldiers and has a huge databank.

42. Note on Immunosorbent Electron Microscopy?

ISEM is a diagnostic technique used primarily in virology. Virus-specific antibodies conjugated to gold particles. These virus antibodies are used to visualize virus particles on electron microscopes. The gold is electron dense • It is seen as a dark shadow against the light background of the specimen field. This Technique used in both transmission or scanning systems. If gold-labelled primary antibodies are not available then anti-IgG–gold conjugated antibodies can be used. These anti-IgG–gold conjugated antibodies can be use with the primary antibody in a double antibody system. Both monoclonal and polyclonal antibodies can be used for ISEM.

43. Note on Variable Number tandem Repeated?

A Variable Number Tandem Repeat (or VNTR) is a location in a genome where a short nucleotide sequence is organized as a tandem repeat. These can be found on many chromosomes, and often show variations in length between individuals. Each variant acts as an inherited allele, allowing them to be used for personal or parental identification.

There are two principal families of VNTRs:

- ***Microsatellites***: Microsatellites, also known as Simple Sequence Repeats (SSRs) or short tandem repeats (STRs), are repeating sequences of 26 base pairs of DNA:
- ***Minisatellites***: • A minisatellites (also referred as VNTR) is a section of DNA that consists of a short series of bases 10–60 base pairs. • Their analysis is useful in genetics and biology research, forensics, and DNA fingerprinting.

44. Note on RFLP/ Procedure of RFLP

• RFLP analyzes the length of the strands of the DNA molecules with repeating base pair patterns. • The basic technique for detecting RFLPs involves fragmenting a sample of DNA by a restriction enzyme, which can recognize and cut DNA wherever a specific short sequence occurs, in a process

known as a ***restriction digest.*** • The resulting DNA fragments are then separated by length through a process known as ***agarose gel electrophoresis***, and transferred to a membrane via the Southern blot procedure. Hybridization of the membrane to a labeled DNA probe then determines the length of the fragments which are complementary to the probe. • An RFLP occurs when the length of a detected fragment varies between individuals. Each fragment length is considered an allele, and can be used in genetic analysis.

45. Southern Blotting Procedure

Southern blot is used to detect the presence of a particular piece of DNA in a sample by a molecular probe. Southern Blotting is named after its inventor, the British Biologist Edwin Southern (1975).

☐ Identify DNA sequence (gene) of interest. ☐ Identified DNA may be a small piece of DNA or a mutation

PROCEDURE:

- DNA is extracted from cells, leukocytes.
- DNA is cleaved into many fragments by restriction enzyme (e.g, BamH1, EcoR1 etc)
- The resulting fragments are separated on the basis of size by electrophoresis.
- The DNA fragments are denatured and transferred to nitrocellulose membrane for analysis.
- The labeled probe is added to the blocked membrane in buffer and incubated for several hours to allow the probe molecules to find their targets.
- Blot is incubated with wash buffers containing NaCl and detergent to wash away excess probe.
- Radioactive probes enable autoradiographic detection.

46. Western Blotting Procedure

☐ A technique used to detect the presence of a specific protein in a complex protein mixture. A molecular technique to study proteins.

PROCEDURE:

1) ***Sample preparation:*** The choice of extraction method depends primarily on the sample and whether the analysis is targeting all the proteins in a cell or only a component from a particular subcellular fraction.
2) ***Gel Electrophoresis:*** Samples are loaded into separate wells. A protein marker is also loaded. The separated protein mixtures are transferred to a solid support for further analysis.
3) ***Blotting (or transfer):*** Transfer can be done in wet or semi-dry conditions. Semi-dry transfer is generally faster. Wet transfer is recommended for large proteins, >100 kD:
4) ***Blocking:*** Blocking is a very important step in the immunodetection phase of Western blotting because it prevents non-specific binding of antibody to the blotting membrane.
5) ***Antibody probing:*** Protein of interest is detected and localized using a specific antibody. Western blotting protocols utilize a non-labeled primary antibody directed against the target protein. A species-specific, labeled secondary antibody directed against the constant region of the primary antibody is then used. The most common antibody label used in Western blots is HRP.
6) ***Detection:*** The signal is detected when HRP is exposed to a substrate solution in the final step of the immunodetection procedure.